

Inspection settings

Adjust the operating threshold to balance missed defects against additional manual inspections.

Minimum probability required to pass as OK



Manual-review margin



Model configuration

MobileNetV2 transfer learning

224 × 224 px

Binary classification

Prototype for academic demonstration.
Not validated for production use.

AI-POWERED MANUFACTURING QUALITY CONTROL

```
<div class="hero-title">
  Nemak Visual Quality Inspection
</div>

<div class="hero-subtitle">
  Upload an image of a cast metal component and determine if it appears acceptable or defective.
  Integrate this into your human-in-the-loop workflow by routing
  parts to a quality inspector.
</div>
```

Test ac...

9...

Defect ...

9...

Defect...

4...

Model ...

M...

INPUT

Upload casting image

Supported formats: JPG, JPEG and PNG.
For best results, use a centred, top-view image similar to the model's training data.

DECISION SUPPORT

Inspection result

The result combines the CNN prediction, the selected operating threshold and the manual-review margin.

 cast_def_0_220.jpeg 
9.5KB

+



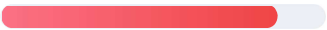
Unloaded component

MODEL
DECISION

```
<div class=
  Defecti
</div>

<div class=
  The com
  product
</div>
```

Defect
confidence



85.0% estimated
probability of defect

Recommended
action: isolate the
component and
perform manual
inspection.

Def...
8..

OK...
1..

> View decision
details

How the prototype supports quality control

STEP 01

Capture

A camera captures a standardised image of the cast component at the inspection point.

STEP 02

Classify

MobileNetV2 analyses visual patterns and estimates the probability that the part is acceptable.

STEP 03

Act

Acceptable parts continue, while suspected or uncertain parts are diverted for human review.

Academic prototype developed for the Neural Networks & Deep Learning Applied to Business module. Predictions should not be used as the sole basis for real manufacturing quality decisions.